

B. Tech Degree III Semester Examination November 2010

SE 304 MANUFACTURING PROCESSES (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) What is "Eutectic System"? Explain.
 (b) Write notes on High Speed Steels.
 (c) Describe the process of Laser beam welding.
 (d) Differentiate DCSP and DCRP in arc welding.
 (e) Explain "Gating System" of Sand Casting Moulds.
 (f) Describe the process of radiographic testing of casting.
 (g) Describe the process of USM.
 (h) Classify Machine tools.

PART B

(4 x 15 = 60)

- II. (a) Describe the process of making pig iron from iron ore with a neat sketch. (8)
 (b) List out the mechanical properties of materials and explain briefly. (7)

OR

- III. (a) List out seven popular alloys of copper and briefly explain each giving the composition and uses. (7)
 (b) Explain the following heat treatment processes of steel (i) Annealing (ii) Case hardening (8)

- IV. (a) What are the different flames employed in oxy-acetylene welding? Draw the flames highlighting the different zones and explain the characteristics and uses of each one. (9)
 (b) Explain friction welding process with an example. (6)

OR

- V. (a) Describe the (GTAW) Gas tungsten arc welding process with a neat sketch. What are merits and limitations of this process. (9)
 (b) Describe the process of cutting metals using oxy-acetylene flame. (6)

- VI. (a) What are the different allowance made while making patterns? Explain. (7)
 (b) Explain the process of cold chamber die casting with a neat sketch. (8)

OR

- VII. (a) Explain the testing procedure of the following properties of moulding sand: (i) Porosity (ii) Clay content (iii) Moisture content. (6)
 (b) Explain the process of investment casting with neat sketches. (9)

- VIII. (a) What are the advantages and drawbacks of hot working? (6)
 (b) With a neat sketch explain the construction and working of a pneumatic hammer. (9)

OR

- IX. (a) What are the different operations that can be performed in a lathe? List and describe any three. (6)
 (b) Explain the quick return mechanism of a shaper with a neat sketch. (9)